Problem 4.29: Step-by-Step Using CG Tables

No angular momentum algebra — just table lookups

Following the Zettili notation and the student's own method

The Problem

Find the ratio of cross sections for:

(a) $\pi^- + p \to K^0 + \Sigma^0$

(b) $\pi^- + p \to K^+ + \Sigma^-$

(c) $\pi^+ + p \to K^+ + \Sigma^+$

First when $I = 3/2$ dominates, then when $I = 1/2$ dominates.

Step 1: Write Down Quark Content

For each particle in the problem, write the quark content:

Initial State	Quarks	Final State	Quarks
p	uud	K^+	$u\bar{s}$
π^-	$d\bar{u}$	K^0	$d\bar{s}$
π^+	$u\bar{d}$	Σ^+	uus
		Σ^0	uds
		Σ^-	dds

Step 2: Assign Isospin Quantum Numbers

Key facts

- u quark: $I = 1/2$, $I_3 = +1/2$
- d quark: $I = 1/2$, $I_3 = -1/2$
- s quark: $I = 0$, $I_3 = 0$ (invisible to isospin)
- Antiquarks flip I_3: $\bar{u}$ has $I_3 = -1/2$, $\bar{d}$ has $I_3 = +1/2$
- I_3 is additive: just add up all the quarks

- Total I comes from the triangle rule, or from the multiplet size: $2I + 1 =$ number of particles in the multiplet

Proton ($p = uud$)

I_3: add up the quarks: $+1/2 + 1/2 + (-1/2) = +1/2$.
Total I: three quarks, each $I = 1/2$. Apply triangle rule twice:

- First two quarks: $1/2 \otimes 1/2 = 0 \oplus 1$
- Then with third: $0 \otimes 1/2 = 1/2$ or $1 \otimes 1/2 = 1/2 \oplus 3/2$
- Possibilities: $I = 1/2$ or $3/2$

The proton is part of a **doublet** (just p and n), so $2I + 1 = 2$, giving $I = 1/2$.

$$\boxed{p = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle} \tag{1}$$

Pion ($\pi^- = d\bar{u}$)

$I_3 = -1/2 + (-1/2) = -1$.
Total I: two light (anti)quarks: $1/2 \otimes 1/2 = 0 \oplus 1$. But $I_3 = -1$ can only fit in $I = 1$ (since $I = 0$ only allows $I_3 = 0$). Stretched state!

$$\boxed{\pi^- = |1, -1\rangle} \tag{2}$$

Similarly: $\pi^+ = |1, +1\rangle$.

Kaons

K^+ is $u\bar{s}$. Only one light quark (u), so $I = 1/2$ automatically.
$I_3(K^+) = +1/2 + 0 = +1/2$. $I_3(K^0) = -1/2 + 0 = -1/2$.

$$\boxed{K^+ = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle, \qquad K^0 = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{3}$$

Sigmas

Σ^+ is uus. Two light quarks: $1/2 \otimes 1/2 = 0 \oplus 1$. $I_3 = +1/2 + 1/2 = +1$. Only $I = 1$ fits. Stretched state!

$$\boxed{\Sigma^+ = |1, +1\rangle, \qquad \Sigma^0 = |1, 0\rangle, \qquad \Sigma^- = |1, -1\rangle} \tag{4}$$

All isospin assignments in one place

Particle	I	I_3
p	$1/2$	$+1/2$
π^-	1	-1
π^+	1	$+1$
K^+	$1/2$	$+1/2$
K^0	$1/2$	$-1/2$
Σ^+	1	$+1$
Σ^0	1	0
Σ^-	1	-1

In practice: look these up in the PDG or Griffiths. The derivations above are just so you understand why.

Step 3: Find the Allowed Total Isospin Channels

Initial state: $\pi + N$

Pion has $I_\pi = 1$, nucleon has $I_N = 1/2$. Apply the triangle rule:

$$|I_\pi - I_N| \leq I \leq I_\pi + I_N \quad \Longrightarrow \quad \frac{1}{2} \leq I \leq \frac{3}{2} \tag{5}$$

So: $I = 1/2$ or $I = 3/2$.

Final state: $K + \Sigma$

Kaon has $I_K = 1/2$, sigma has $I_\Sigma = 1$. Triangle rule:

$$|I_K - I_\Sigma| \leq I \leq I_K + I_\Sigma \quad \Longrightarrow \quad \frac{1}{2} \leq I \leq \frac{3}{2} \tag{6}$$

Same two channels: $I = 1/2$ or $I = 3/2$.

Two amplitudes govern everything

The strong force conserves total isospin, so scattering can only happen within each channel independently. There are two reduced amplitudes:

- $\mathcal{A}_{3/2}$ = amplitude for the $I = 3/2$ channel
- $\mathcal{A}_{1/2}$ = amplitude for the $I = 1/2$ channel

All three reactions share these same two numbers. The only difference between reactions is how the CG coefficients weight them.

Step 4: Narrow Down Using I_3 Conservation

I_3 is additive and conserved. For each reaction, compute the total I_3 and see which $|I, I_3\rangle$ states are allowed.

Reactions (a) and (b): $\pi^- + p$ initial state

$$I_{3,\text{total}} = I_{3,\pi^-} + I_{3,p} = (-1) + (+\tfrac{1}{2}) = -\frac{1}{2} \tag{7}$$

Which total-isospin states have $I_3 = -1/2$?

- $I = 3/2$: allowed I_3 values are $-3/2, -1/2, +1/2, +3/2$. **Yes**, $-1/2$ is in the list.
- $I = 1/2$: allowed I_3 values are $-1/2, +1/2$. **Yes**, $-1/2$ is in the list.

Both channels contribute. We need: $|3/2, -1/2\rangle$ and $|1/2, -1/2\rangle$.

Reaction (c): $\pi^+ + p$ initial state

$$I_{3,\text{total}} = I_{3,\pi^+} + I_{3,p} = (+1) + (+\tfrac{1}{2}) = +\frac{3}{2} \tag{8}$$

Which states have $I_3 = +3/2$?

- $I = 3/2$: **Yes**, $+3/2$ is allowed.
- $I = 1/2$: maximum I_3 is $+1/2$. **No!** $+3/2$ doesn't fit.

Only one channel contributes. π^+p is **pure** $I = 3/2$.

This is the key observation!

π^+p has $I_3 = +3/2$, which is too big for $I = 1/2$. So it has **zero overlap** with the $I = 1/2$ channel. When $I = 1/2$ dominates, reaction (c) is completely forbidden!

Step 5: Decompose Initial States Using the CG Table

The Zettili formula (Eq. 7.159)

To express a total-isospin state $|j, m\rangle$ in terms of product states:

$$|j, m\rangle = \sum_{m_1} \sum_{m_2} \langle j_1, j_2;\, m_1, m_2 \,|\, j, m\rangle \; |j_1, j_2;\, m_1, m_2\rangle \tag{9}$$

Or equivalently, to express a product state in terms of total-isospin states (read the CG table by **column**):

$$|j_1, j_2;\, m_1, m_2\rangle = \sum_{j} \langle j_1, j_2;\, m_1, m_2 \,|\, j, m\rangle \; |j, m\rangle \tag{10}$$

where $m = m_1 + m_2$ is fixed.

How to read the CG table

The table is organized as $j_1 \times j_2$ (bigger isospin first). Each entry is a number with **a square root understood over it**. So "2/3" in the table means $\sqrt{2/3}$, and "$-1/3$" means $-\sqrt{1/3}$.
To decompose a product state: find the **column** for your m_1, m_2 values, then read down to get the coefficient for each $|J, M\rangle$ row.

$|\pi^- p\rangle$ decomposition

We have $|\pi^- p\rangle = |I_\pi = 1,\ I_p = 1/2;\ m_1 = -1,\ m_2 = +1/2\rangle$.

This matches the $1 \times 1/2$ CG table. Go to the column $m_1 = -1$, $m_2 = +1/2$ (which has $M = -1/2$). Read down:

Row in table	Table entry	Actual coefficient
$\|3/2,\ -1/2\rangle$	$1/3$	$\sqrt{1/3}$
$\|1/2,\ -1/2\rangle$	$-2/3$	$-\sqrt{2/3}$

Therefore:

$$\boxed{|\pi^- p\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{11}$$

Normalization check

$\left(\sqrt{1/3}\right)^2 + \left(-\sqrt{2/3}\right)^2 = 1/3 + 2/3 = 1$ ✓

$|\pi^+ p\rangle$ decomposition

We have $|\pi^+ p\rangle = |1, 1/2;\ +1, +1/2\rangle$, with $M = +3/2$.

Go to the $1 \times 1/2$ table, column $m_1 = +1$, $m_2 = +1/2$. The only nonzero entry is:

Row	Table entry	Actual coefficient
$\|3/2,\ +3/2\rangle$	1	$\sqrt{1} = 1$

$$\boxed{|\pi^+ p\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle} \tag{12}$$

Pure $I = 3/2$. Stretched state — only one entry in the column.

Step 6: Decompose Final States Using the CG Table

The final states have the kaon ($I = 1/2$) and sigma ($I = 1$). Since we're putting the $I = 1/2$ particle first, we're doing $1/2 \otimes 1$ instead of $1 \otimes 1/2$.

Coupling order phase convention

The CG table is printed for $1 \times 1/2$ (bigger isospin first). When we swap to $1/2 \times 1$ (kaon first), we use:

$$\langle j_2, j_1;\, m_2, m_1 \,|\, J, M\rangle = (-1)^{j_1+j_2-J}\, \langle j_1, j_2;\, m_1, m_2 \,|\, J, M\rangle \tag{13}$$

With $j_1 = 1$, $j_2 = 1/2$:

- $J = 3/2$: phase $= (-1)^{1+1/2-3/2} = (-1)^0 = +1$ (no change)
- $J = 1/2$: phase $= (-1)^{1+1/2-1/2} = (-1)^1 = -1$ (flip sign)

Recipe: Look up the $1 \times 1/2$ table as usual, but multiply every $I = 1/2$ coefficient by (-1).

$|K^0\Sigma^0\rangle$ decomposition (reaction a final state)

In isospin: $|K^0\Sigma^0\rangle = |1/2, 1;\ -1/2, 0\rangle$. Total $I_3 = -1/2 + 0 = -1/2$.

Step 1: Find it in the $1 \times 1/2$ table. We need to swap the order to read the table. Our $m_2 = -1/2$ (kaon) becomes m_2 in the table, and our $m_1 = 0$ (sigma) becomes m_1 in the table. So look up the column $m_1 = 0$, $m_2 = -1/2$:

Row	Table entry	Phase	Final coefficient
$\|3/2,\ -1/2\rangle$	$\sqrt{2/3}$	$\times(+1)$	$\sqrt{2/3}$
$\|1/2,\ -1/2\rangle$	$\sqrt{1/3}$	$\times(-1)$	$-\sqrt{1/3}$

Now we also need the column $m_1 = -1$, $m_2 = +1/2$ (since $(-1) + 1/2 = -1/2$ also works):

Row	Table entry	Phase	Final coefficient
$\|3/2,\ -1/2\rangle$	$\sqrt{1/3}$	$\times(+1)$	$\sqrt{1/3}$
$\|1/2,\ -1/2\rangle$	$-\sqrt{2/3}$	$\times(-1)$	$\sqrt{2/3}$

But wait — we only need the column that matches our specific m_1, m_2. For $K^0\Sigma^0$: $m_K = -1/2$, $m_\Sigma = 0$. In the *swapped* table that's $m_1 = 0$ (sigma), $m_2 = -1/2$ (kaon). So we use the **first** table above.

$$\boxed{|K^0\Sigma^0\rangle = \sqrt{\frac{2}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{14}$$

Normalization check

$2/3 + 1/3 = 1$ ✓

$|K^+\Sigma^-\rangle$ decomposition (reaction b final state)

In isospin: $|K^+\Sigma^-\rangle = |1/2, 1;\ +1/2, -1\rangle$. Total $I_3 = +1/2 + (-1) = -1/2$.

In the swapped table: $m_1 = -1$ (sigma), $m_2 = +1/2$ (kaon). Look up column $m_1 = -1$, $m_2 = +1/2$:

Row	Table entry	Phase	Final coefficient
$\|3/2, -1/2\rangle$	$\sqrt{1/3}$	$\times(+1)$	$\sqrt{1/3}$
$\|1/2, -1/2\rangle$	$-\sqrt{2/3}$	$\times(-1)$	$\sqrt{2/3}$

$$\boxed{|K^+\Sigma^-\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{15}$$

Normalization check

$1/3 + 2/3 = 1$ ✓

$|K^+\Sigma^+\rangle$ decomposition (reaction c final state)

In isospin: $|K^+\Sigma^+\rangle = |1/2, 1;\ +1/2, +1\rangle$. Total $I_3 = +1/2 + 1 = +3/2$.

Only $I = 3/2$ can have $I_3 = +3/2$. Stretched state!

$$\boxed{|K^+\Sigma^+\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle} \tag{16}$$

Summary of all decompositions

Initial states:

$$|\pi^- p\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{I-1}$$

$$|\pi^+ p\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle \tag{I-2}$$

Final states (kaon-first coupling order):

$$|K^0\Sigma^0\rangle = \sqrt{\frac{2}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{F-1}$$

$$|K^+\Sigma^-\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{F-2}$$

$$|K^+\Sigma^+\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle \tag{F-3}$$

Step 7: Compute the Amplitudes

The amplitude formula

The physical amplitude for a reaction is:

$$\mathcal{M} = \langle \text{final} | \, T \, | \text{initial} \rangle = \sum_I (\text{final CG for } I) \times \mathcal{A}_I \times (\text{initial CG for } I) \tag{17}$$

The strong force only connects states with the **same** total I, so we multiply the CG coefficients for each channel and sum.

Reaction (a): $\pi^- p \to K^0 \Sigma^0$

Read off the CG coefficients from (I-1) and (F-1):

Channel	Initial coeff.	Final coeff.	Product
$I = 3/2$	$\sqrt{1/3}$	$\sqrt{2/3}$	$\sqrt{1/3} \times \sqrt{2/3} = \sqrt{2}/3$
$I = 1/2$	$-\sqrt{2/3}$	$-\sqrt{1/3}$	$(-\sqrt{2/3}) \times (-\sqrt{1/3}) = \sqrt{2}/3$

Add them up:

$$\mathcal{M}_a = \frac{\sqrt{2}}{3} \mathcal{A}_{3/2} + \frac{\sqrt{2}}{3} \mathcal{A}_{1/2} \tag{18}$$

$$\boxed{\mathcal{M}_a = \frac{\sqrt{2}}{3} \left(\mathcal{A}_{3/2} + \mathcal{A}_{1/2} \right)} \tag{19}$$

Let's trace through this carefully

$I = 3/2$ channel: initial state has coefficient $\sqrt{1/3}$ in $|3/2, -1/2\rangle$, final state has coefficient $\sqrt{2/3}$ in $|3/2, -1/2\rangle$. Multiply: $\sqrt{1/3} \times \sqrt{2/3} = \sqrt{2/9} = \sqrt{2}/3$.
$I = 1/2$ channel: initial has $-\sqrt{2/3}$, final has $-\sqrt{1/3}$. Multiply: $(-\sqrt{2/3})(-\sqrt{1/3}) = +\sqrt{2}/3$.
Both channels give $+\sqrt{2}/3$, so $\mathcal{A}_{3/2}$ and $\mathcal{A}_{1/2}$ add with the same sign. ✓

Reaction (b): $\pi^- p \to K^+ \Sigma^-$

Read off CG coefficients from (I-1) and (F-2):

Channel	Initial coeff.	Final coeff.	Product
$I = 3/2$	$\sqrt{1/3}$	$\sqrt{1/3}$	$\sqrt{1/3} \times \sqrt{1/3} = 1/3$
$I = 1/2$	$-\sqrt{2/3}$	$\sqrt{2/3}$	$(-\sqrt{2/3}) \times (\sqrt{2/3}) = -2/3$

Add them up:

$$\mathcal{M}_b = \frac{1}{3}\,\mathcal{A}_{3/2} + \left(-\frac{2}{3}\right)\mathcal{A}_{1/2} \tag{20}$$

$$\boxed{\mathcal{M}_b = \frac{1}{3}\left(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right)} \tag{21}$$

Reaction (c): $\pi^+ p \to K^+\Sigma^+$

From (I-2) and (F-3): both states are pure $|3/2, +3/2\rangle$ with coefficient 1.

Channel	Initial coeff.	Final coeff.	Product
$I = 3/2$	1	1	$1 \times 1 = 1$
$I = 1/2$	0	0	0

$$\boxed{\mathcal{M}_c = \mathcal{A}_{3/2}} \tag{22}$$

The simplest reaction

Both initial and final states are pure $I = 3/2$ (stretched states with $I_3 = +3/2$). The amplitude is just $\mathcal{A}_{3/2}$ with no mixing, no CG coefficients to worry about (they're both 1).

Step 8: Compute Cross-Section Ratios

Cross section is proportional to $|\mathcal{M}|^2$:

$$\sigma_a \propto |\mathcal{M}_a|^2 = \frac{2}{9}\left|\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right|^2 \tag{23}$$

$$\sigma_b \propto |\mathcal{M}_b|^2 = \frac{1}{9}\left|\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right|^2 \tag{24}$$

$$\sigma_c \propto |\mathcal{M}_c|^2 = \left|\mathcal{A}_{3/2}\right|^2 \tag{25}$$

Case 1: $I = 3/2$ dominates ($\mathcal{A}_{1/2} \approx 0$)

Set $\mathcal{A}_{1/2} = 0$:

$$\sigma_a \propto \frac{2}{9}|\mathcal{A}_{3/2}|^2 \tag{26}$$

$$\sigma_b \propto \frac{1}{9}|\mathcal{A}_{3/2}|^2 \tag{27}$$

$$\sigma_c \propto |\mathcal{A}_{3/2}|^2 \tag{28}$$

The $|\mathcal{A}_{3/2}|^2$ cancels in the ratio. Multiply by 9 to clear fractions:

$$\boxed{\sigma_a : \sigma_b : \sigma_c = 2 : 1 : 9} \tag{29}$$

Case 2: $I = 1/2$ dominates ($\mathcal{A}_{3/2} \approx 0$)

Set $\mathcal{A}_{3/2} = 0$:

$$\sigma_a \propto \frac{2}{9}|\mathcal{A}_{1/2}|^2 \tag{30}$$

$$\sigma_b \propto \frac{1}{9} \times 4|\mathcal{A}_{1/2}|^2 = \frac{4}{9}|\mathcal{A}_{1/2}|^2 \tag{31}$$

$$\sigma_c \propto 0 \tag{32}$$

Multiply by $9/(2|\mathcal{A}_{1/2}|^2)$:

$$\boxed{\sigma_a : \sigma_b : \sigma_c = 1 : 2 : 0} \tag{33}$$

$\sigma_c = 0$: the most striking prediction

The reaction $\pi^+ p \to K^+\Sigma^+$ is **completely forbidden** when $I = 1/2$ dominates. Why? Because $|\pi^+ p\rangle$ is pure $I = 3/2$ (it has $I_3 = +3/2$, which is too big for $I = 1/2$). If the $I = 3/2$ amplitude is turned off, there is simply nothing for $\pi^+ p$ to scatter through. The quantum numbers don't match.

Step 9: Final Summary

	Amplitude $\mathcal{M}$	$I = 3/2$ dom.	$I = 1/2$ dom.
(a) $\pi^- p \to K^0\Sigma^0$	$\frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2})$	2	1
(b) $\pi^- p \to K^+\Sigma^-$	$\frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2})$	1	2
(c) $\pi^+ p \to K^+\Sigma^+$	$\mathcal{A}_{3/2}$	9	0

Consistency check: amplitude relation

The three amplitudes satisfy:

$$\mathcal{M}_c = \sqrt{2}\,\mathcal{M}_a + \mathcal{M}_b \tag{34}$$

Check:

$$\begin{aligned}\sqrt{2}\,\mathcal{M}_a + \mathcal{M}_b &= \sqrt{2} \cdot \frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}) + \frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}) \\ &= \frac{2}{3}\mathcal{A}_{3/2} + \frac{2}{3}\mathcal{A}_{1/2} + \frac{1}{3}\mathcal{A}_{3/2} - \frac{2}{3}\mathcal{A}_{1/2} \\ &= \mathcal{A}_{3/2} = \mathcal{M}_c \quad \checkmark\end{aligned} \tag{35}$$

This identity holds for any $\mathcal{A}_{3/2}$ and $\mathcal{A}_{1/2}$ — it's a model-independent consequence of isospin symmetry.

The recipe you just followed

1. **Quark content** → identify particles
2. **Isospin assignments** → write each particle as $|I, I_3\rangle$ (use PDG or stretched state argument)
3. **Triangle rule** → find allowed total isospin channels
4. I_3 **conservation** → narrow down which $|I, I_3\rangle$ states you need
5. **CG table lookup** → decompose initial and final product states (apply coupling-order phase if needed)
6. **Multiply CG coefficients** → get amplitude for each reaction
7. **Square and take ratios** → cross-section ratios

No angular momentum algebra required. Just table lookups and multiplication.